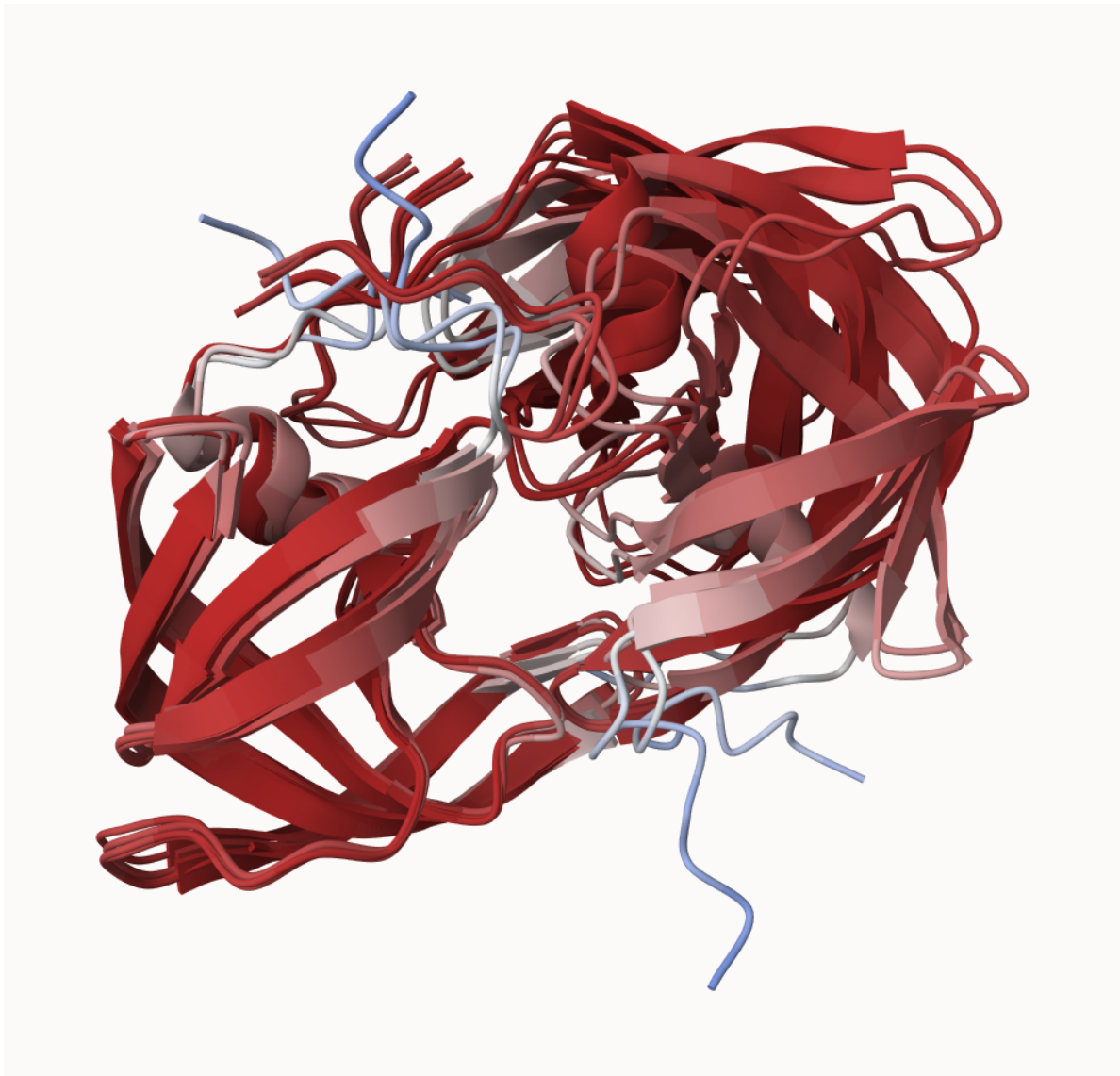


class11

Alishba Qadri

Figure 18:



8. Custom analysis of resulting models

```
# Change this for YOUR results dir name  
results_dir <- "proteinsearch_23119"
```

```
# File names for all PDB models  
pdb_files <- list.files(path=results_dir,
```

```

        pattern="*.pdb",
        full.names = TRUE)

# Print our PDB file names
basename(pdb_files)

```

```

[1] "proteinsearch_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb"
[2] "proteinsearch_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb"
[3] "proteinsearch_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb"
[4] "proteinsearch_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb"
[5] "proteinsearch_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"

```

```

# load bio3d
library(bio3d)

# Read all data from Models
# and superpose/fit coords
pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")

```

Reading PDB files:

```

proteinsearch_23119/proteinsearch_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb
proteinsearch_23119/proteinsearch_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb
proteinsearch_23119/proteinsearch_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb
proteinsearch_23119/proteinsearch_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb
proteinsearch_23119/proteinsearch_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb
.....

```

Extracting sequences

```

pdb/seq: 1   name: proteinsearch_23119/proteinsearch_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb
pdb/seq: 2   name: proteinsearch_23119/proteinsearch_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb
pdb/seq: 3   name: proteinsearch_23119/proteinsearch_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb
pdb/seq: 4   name: proteinsearch_23119/proteinsearch_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb
pdb/seq: 5   name: proteinsearch_23119/proteinsearch_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb

```

```

# quick overview of model sequences
pdbs

```

```

[Truncated_Name:1]proteinsea      1          .          .          .          .          50
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI

```

```

[Truncated_Name:2]proteinsea PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKPMIGGI
[Truncated_Name:3]proteinsea PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKPMIGGI
[Truncated_Name:4]proteinsea PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKPMIGGI
[Truncated_Name:5]proteinsea PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKPMIGGI
*****
1 . . . . 50

51 . . . . 100
[Truncated_Name:1]proteinsea GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:2]proteinsea GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:3]proteinsea GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:4]proteinsea GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:5]proteinsea GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
*****
51 . . . . 100

101 . . . . 150
[Truncated_Name:1]proteinsea QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKPMIGGIG
[Truncated_Name:2]proteinsea QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKPMIGGIG
[Truncated_Name:3]proteinsea QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKPMIGGIG
[Truncated_Name:4]proteinsea QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKPMIGGIG
[Truncated_Name:5]proteinsea QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKPMIGGIG
*****
101 . . . . 150

151 . . . . 198
[Truncated_Name:1]proteinsea GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:2]proteinsea GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:3]proteinsea GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:4]proteinsea GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:5]proteinsea GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
*****
151 . . . . 198

```

Call:

```
pdbaln(files = pdb_files, fit = TRUE, exefile = "msa")
```

Class:

```
pdb, fasta
```

Alignment dimensions:

```
5 sequence rows; 198 position columns (198 non-gap, 0 gap)
```

```
+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

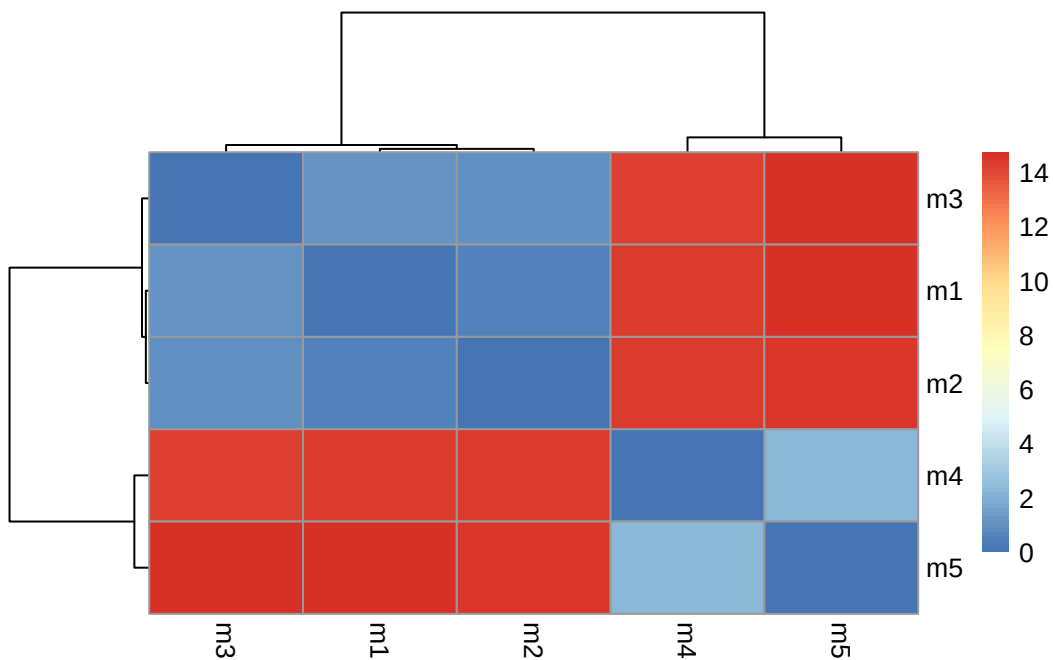
```
# use rmsd() function to calculate RMSD between all paris models  
rd <- rmsd(pdbbs, fit=T)
```

Warning in rmsd(pdbbs, fit = T): No indices provided, using the 198 non NA positions

```
range(rd)
```

```
[1] 0.000 14.754
```

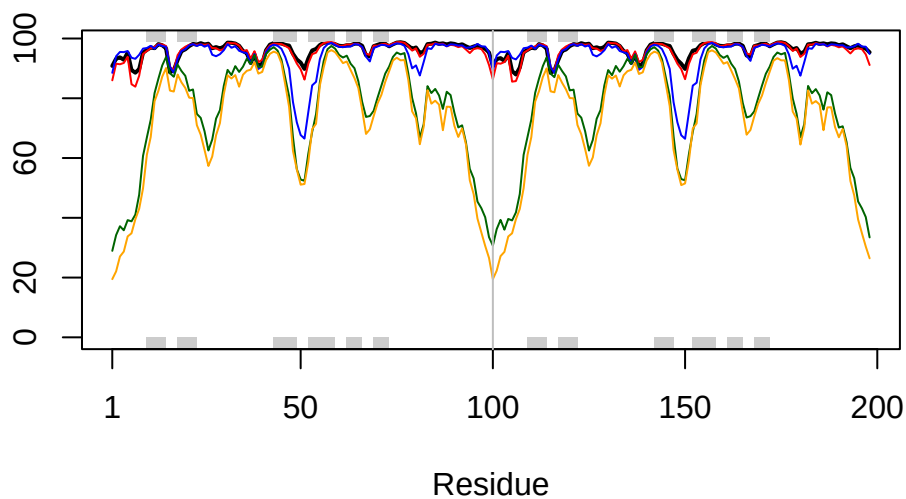
```
# load pheatmap  
library(pheatmap)  
# draw a heatmap of RMSD matrix values  
colnames(rd) <- paste0("m",1:5)  
rownames(rd) <- paste0("m",1:5)  
pheatmap(rd)
```



```
# Read a reference PDB structure  
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
# plot pLDDT values across all models
plotb3(pdbb$b[1,], typ="l", lwd=2, sse=pdbb)
points(pdbb$b[2,], typ="l", col="red")
points(pdbb$b[3,], typ="l", col="blue")
points(pdbb$b[4,], typ="l", col="darkgreen")
points(pdbb$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```



```
# find most consistent "rigid core" common across all models
core <- core.find(pdbb)
```

```
core size 197 of 198  vol = 9836.196
core size 196 of 198  vol = 6800.793
core size 195 of 198  vol = 1322.091
core size 194 of 198  vol = 1032.249
core size 193 of 198  vol = 944.133
core size 192 of 198  vol = 894.048
core size 191 of 198  vol = 829.724
core size 190 of 198  vol = 766.628
core size 189 of 198  vol = 728.681
```

core size 188 of 198	vol = 693.317
core size 187 of 198	vol = 656.166
core size 186 of 198	vol = 622.063
core size 185 of 198	vol = 586.601
core size 184 of 198	vol = 565.358
core size 183 of 198	vol = 532.747
core size 182 of 198	vol = 511.421
core size 181 of 198	vol = 489.077
core size 180 of 198	vol = 468.599
core size 179 of 198	vol = 449.222
core size 178 of 198	vol = 433.122
core size 177 of 198	vol = 418.651
core size 176 of 198	vol = 405.069
core size 175 of 198	vol = 391.855
core size 174 of 198	vol = 381.006
core size 173 of 198	vol = 371.54
core size 172 of 198	vol = 355.539
core size 171 of 198	vol = 345.156
core size 170 of 198	vol = 335.982
core size 169 of 198	vol = 325.275
core size 168 of 198	vol = 313.603
core size 167 of 198	vol = 302.908
core size 166 of 198	vol = 293.343
core size 165 of 198	vol = 284.547
core size 164 of 198	vol = 277.932
core size 163 of 198	vol = 269.592
core size 162 of 198	vol = 258.167
core size 161 of 198	vol = 246.856
core size 160 of 198	vol = 239.073
core size 159 of 198	vol = 234.174
core size 158 of 198	vol = 226.309
core size 157 of 198	vol = 221.212
core size 156 of 198	vol = 214.951
core size 155 of 198	vol = 206.027
core size 154 of 198	vol = 197.301
core size 153 of 198	vol = 190.889
core size 152 of 198	vol = 185.403
core size 151 of 198	vol = 179.154
core size 150 of 198	vol = 174.709
core size 149 of 198	vol = 167.269
core size 148 of 198	vol = 160.866
core size 147 of 198	vol = 156.173
core size 146 of 198	vol = 148.395

core size 145 of 198	vol = 143.389
core size 144 of 198	vol = 138.428
core size 143 of 198	vol = 132.99
core size 142 of 198	vol = 126.924
core size 141 of 198	vol = 121.257
core size 140 of 198	vol = 116.407
core size 139 of 198	vol = 112.241
core size 138 of 198	vol = 107.813
core size 137 of 198	vol = 104.807
core size 136 of 198	vol = 100.968
core size 135 of 198	vol = 97.21
core size 134 of 198	vol = 92.54
core size 133 of 198	vol = 87.964
core size 132 of 198	vol = 85.621
core size 131 of 198	vol = 81.514
core size 130 of 198	vol = 77.627
core size 129 of 198	vol = 74.881
core size 128 of 198	vol = 72.642
core size 127 of 198	vol = 70.352
core size 126 of 198	vol = 68.627
core size 125 of 198	vol = 66.349
core size 124 of 198	vol = 64.019
core size 123 of 198	vol = 61.763
core size 122 of 198	vol = 58.758
core size 121 of 198	vol = 56.354
core size 120 of 198	vol = 54.129
core size 119 of 198	vol = 51.611
core size 118 of 198	vol = 49.502
core size 117 of 198	vol = 48.045
core size 116 of 198	vol = 46.489
core size 115 of 198	vol = 44.599
core size 114 of 198	vol = 43.185
core size 113 of 198	vol = 40.978
core size 112 of 198	vol = 39.029
core size 111 of 198	vol = 36.363
core size 110 of 198	vol = 33.953
core size 109 of 198	vol = 31.343
core size 108 of 198	vol = 29.304
core size 107 of 198	vol = 27.188
core size 106 of 198	vol = 25.655
core size 105 of 198	vol = 24.03
core size 104 of 198	vol = 22.504
core size 103 of 198	vol = 20.906


```

core size 102 of 198  vol = 19.803
core size 101 of 198  vol = 18.215
core size 100 of 198  vol = 15.652
core size 99 of 198   vol = 14.762
core size 98 of 198   vol = 11.574
core size 97 of 198   vol = 9.391
core size 96 of 198   vol = 7.327
core size 95 of 198   vol = 6.123
core size 94 of 198   vol = 5.592
core size 93 of 198   vol = 4.689
core size 92 of 198   vol = 3.66
core size 91 of 198   vol = 2.779
core size 90 of 198   vol = 2.147
core size 89 of 198   vol = 1.725
core size 88 of 198   vol = 1.158
core size 87 of 198   vol = 0.88
core size 86 of 198   vol = 0.682
core size 85 of 198   vol = 0.527
core size 84 of 198   vol = 0.37
FINISHED: Min vol ( 0.5 ) reached

```

```

# use the identified core atom positions as a basis for more suitable superposition
core.inds <- print(core, vol=0.5)

```

```

# 85 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     9  49     41
2    52  95     44

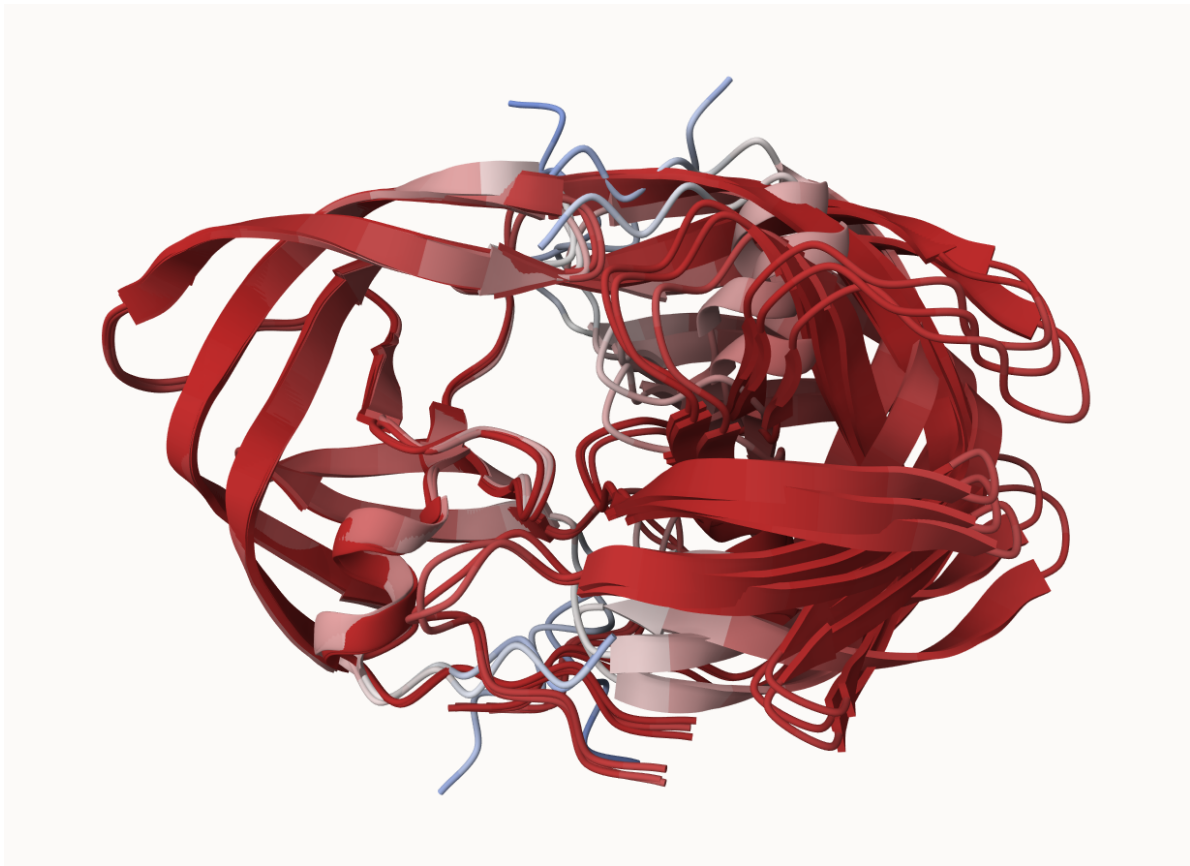
```

```

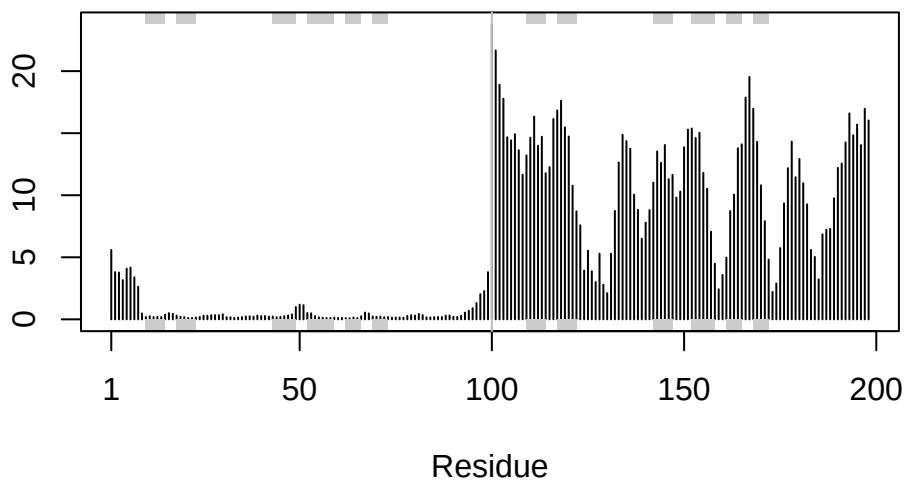
# add superposed coordinates to corefit_structures file
xyz <- pdbfit(pdb, core.inds, outpath="corefit_structures")

```

Figure 19:



```
# examine RMSF between positions of the structure  
rf <- rmsf(xyz)  
  
plotb3(rf, sse=pdb)  
abline(v=100, col="gray", ylab="RMSF")
```



Predicted Alignment Error for domains

```
# load jsonlite
library(jsonlite)

# Listing of all PAE JSON files
pae_files <- list.files(path=results_dir,
                        pattern=".*model.*\\.json",
                        full.names = TRUE)

# read the 1st and 5th files
pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)

attributes(pae1)
```

```
$names
[1] "plddt" "max_pae" "pae" "ptm" "iptm"
```

```
# Per-residue pLDDT scores
# same as B-factor of PDB..
head(pae1$plddt)
```

```
[1] 90.81 93.25 93.69 92.88 95.25 89.44
```

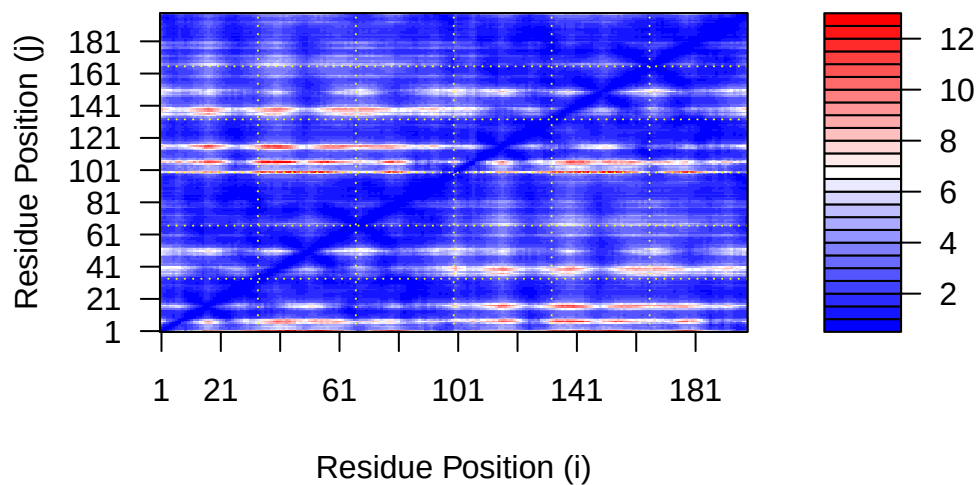
```
# find maximum PAE values for model 1
pae1$max_pae
```

```
[1] 12.84375
```

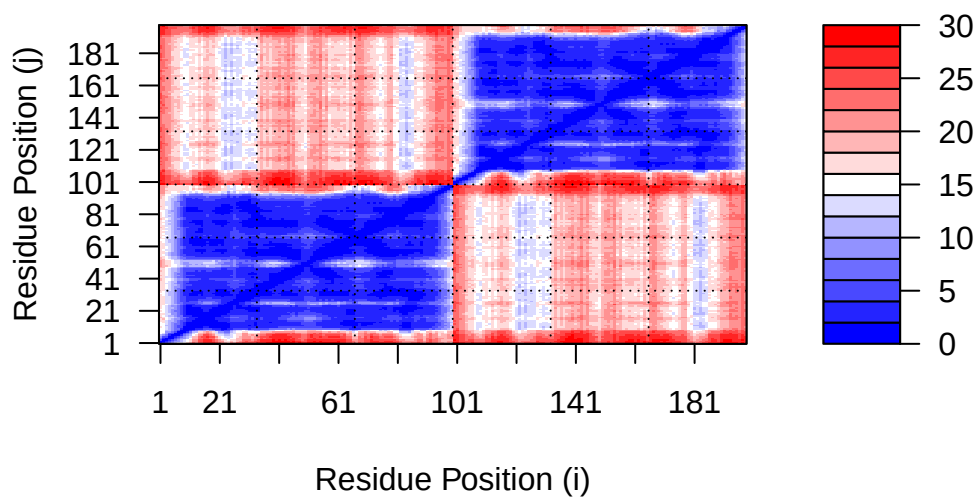
```
# find maximum PAE values for model 5
pae5$max_pae
```

```
[1] 29.59375
```

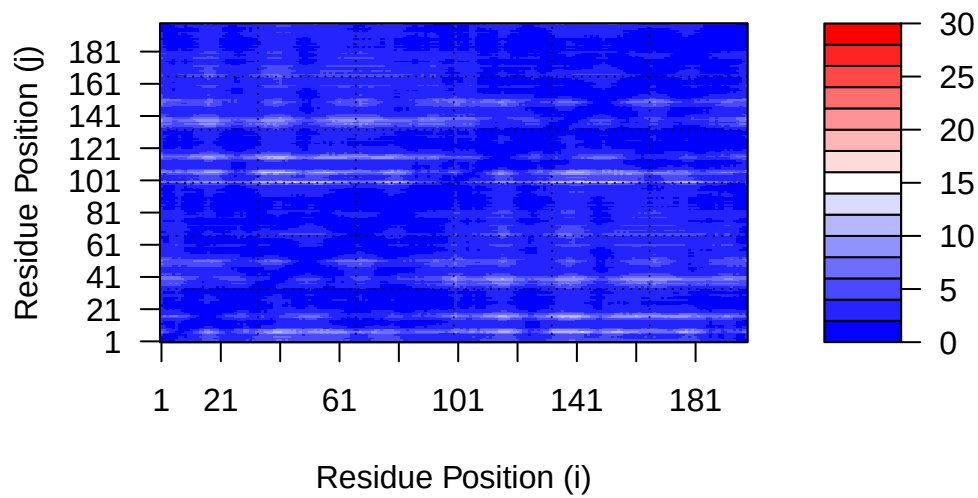
```
# plot the PAE scores of model 1 using Bio3D
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)")
```



```
# plot the PAE scores of model 5 using Bio3D
plot.dmat(pae5$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



```
# plot PAE scores of model 1 using same data range as model 5 plot
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



Residue conservation from alignment file

```
# find file ending in .a3m
aln_file <- list.files(path=results_dir,
                      pattern=".a3m$",
                      full.names = TRUE)

aln_file
```

```
[1] "proteinsearch_23119/proteinsearch_23119.a3m"
```

```
# load sequence into R
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

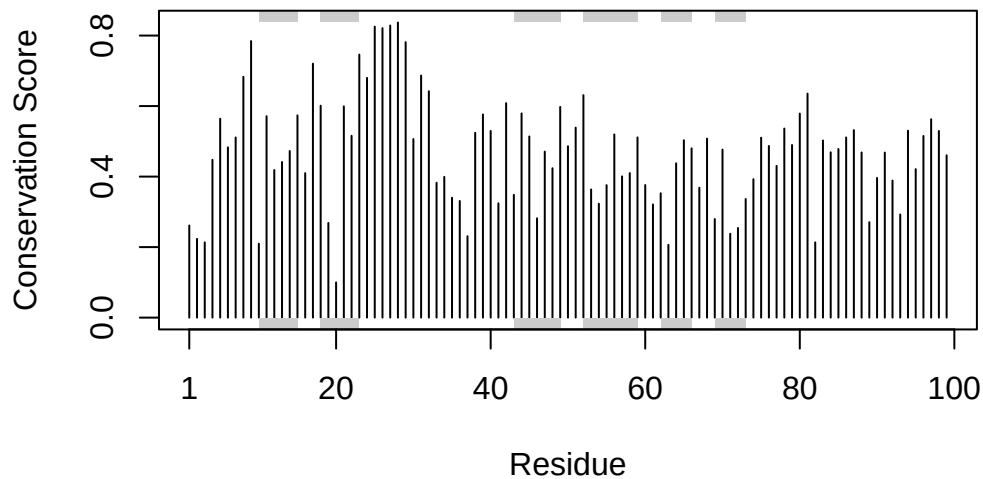
```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

```
# find number of sequences in alignment
dim(aln$ali)
```

```
[1] 5397 132
```

```
# score residue conservation in alignment using conserv() function
sim <- conserv(aln)
```

```
# create conservation plot
plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```



```
# generate a consensus sequence with high cutoff value
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
[1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-"
[37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

```
# map conservation score to Occupancy score of PDB file for viewing in Mol*  
m1.pdb <- read.pdb(pdb_files[1])  
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)  
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```

Figure 20:

